Dermalytics: Skin Cancer Detection Using Convolutional Neural Network (CNN)

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A capstone project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering



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Declaration

We, **Samia Jahan Mariam**, **Khandakar Shatu Moni**, and **Md. Sazzad Hossain** hereby, declare that this capstone project is the result of our own work conducted under the supervision of **Dr. Md. Mostofa Kamal Rasel**, Associate Professor Department of Computer Science and Engineering, East West University. This project has not been submitted elsewhere for any degree or purpose. We further confirm that our research received no personal or financial support from any source.

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Letter of Acceptance

The capstone project titled "Dermalytics: Skin Cancer Detection Using Convolutional Neural Network (CNN)", submitted by Samia Jahan Mariam, Khandakar Shatu Moni, and Md. Sazzad Hossain from the Department of Computer Science and Engineering, has been examined. Upon the examination committee's recommendation, and approve the project and report, as they meet the requirements for partial fulfillment of the Bachelor of Science in Computer Science and Engineering degree on (11.12.2025).

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Acknowledgment

Our study of 'Dermalytics: Skin Cancer Detection Using Convolutional Neural Network (CNN)' has been a great pleasure. Throughout this project, We have acquired the necessary knowledge and experience.

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Finally, we would like to thank all those people who have offered friendship and support related to my academic endeavors; they will be acknowledged more appropriately in the future and will be held in our hearts.

Samia Jahan Mariam
December 2025

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Abstract

The skin cancer as one of the most common types of cancer in the world means that early detection of this disease is critical for effective treatment. In this study, we investigate how deep learning can be applied to classify skin lesions via automated classification methods. More specifically, we examined multiple leading CNN architectures with transfer learning; ResNet, EfficientNet, MobileNet, DenseNet, Inception and Xception. To train these models, we performed preprocessing and augmentation on each dataset. Of all the models tested, Xception achieved the highest classification accuracy (77.56%), precision (85%), recall (95%), and F1 score (90%). This shows that deep learning can accurately differentiate between many types of skin lesions and provides dermatologists with the ability to utilize AI-based tools to accurately detect them.

Keywords: Skin Cancer detection, Skin lesion classification, CNN, Deep Learning, Transfer Learning, Inception, Computer-aided diagnosis

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Chapter 1
Introduction

1.1 Background

Skin disorders are considered a big health issue all over the world and every year, millions of people go through these conditions [1]. Skin includes a broad range of skin diseases, from benign infections and inflammatory skin diseases to malignant skin cancers, with melanoma being the deadliest type of skin disease, followed by basal cell carcinoma and squamous cell carcinoma [2][3]. The category of malignancies (skin cancers) has a very high growth rate and is more likely to cause death during later-stage diagnosis than other types of malignancies because it has such a high percentage of patients that progress to the later stage of the disease [3]. Over time, skin cancer has gradually increased in number due to various factors including the sun, pollution, and genetics [3]. The conventional methods of diagnosis consist of visual exams and taking skin samples for lab analysis, which are very slow and subjective, depending a lot on the skills and knowledge of the dermatologist [4]. Thus, it creates major problems especially in rural or low-income areas where specialized medical assistance is scarcely available [4].

To solve this problem, modern technologies like machine learning (ML) and deep learning (DL) have great potential [5]. These methods can automatically analyze dermoscopic and clinical images of skin lesions. They are trained by machine

learning algorithms (MLA) using images of skin conditions including melanoma, actinic keratosis, vascular lesions, dermatofibromas, nevi, eczema, psoriasis, vitiligo, and fungal infections to identify patterns associated with those conditions. CNNs are a type of deep learning model that perform well in identifying the aforementioned pattern features, such as colour, texture, and shape that are often difficult for human beings to detect visually [5][6].

In this study, we propose Dermalytics, an intelligent computer-aided diagnostic (CAD) system. Through the use of convolutional neural networks (CNNs) and transfer learning, dermalytics helps to identify and classify skin lesions with high precision and speed [5]. The aim of dermalytics is to reduce mistakes, speed up the process of identifying and diagnosing skin lesions, and support dermatologists in their diagnosis of patients. Dermalytics will enable both physicians and patients to utilize the trained model on both mobile and web devices to access dermatology services instantaneously through teledermatology; this will allow for better education and improved clinical decision making through faster access to the most appropriate treatment [6].

1.2 Problem Statement and Analysis

We discuss six main challenges within skin lesion classification. First due to class imbalance within which the malignant lesions are comparatively rare with respect to benign lesions causing a greatly unbalanced dataset. Second due to the visual similarity between different types of lesions which often made it difficult to differentiate and classify them with absolute certainty. Third variability in image quality and skin tones impacted the strength and capacity of the model to generalize that required methods to handle such variation. Fourth, selecting the optimal CNN architecture from several that are already available was a critical issue to attain maximum performance. Fifth, the powerful computational capabilities required to

train deep learning models imposed practical limitations particularly regarding processing power and time. Finally having consistent input images, that is to say ensuring that they are different, properly positioned and of the same size was necessary for stable testing and evaluation of the effectiveness of the model.

The aim of this project is to construct an accurate system for classifying skin lesions, paying special attention to detecting the uncommon malignant ones from among the common benign lesions. Among the major challenges in this task is having a grossly imbalanced class population in that malignant instances are comparatively few compared to benign lesions. Strategies for data augmentation and stratified splitting mitigate this, enabling the model to learn effectively even with limited malignant examples.

The other challenge is the visual appearance similarity across different types of lesions, making their identification difficult. We overcome this challenge by performing transfer learning from pre-trained Convolutional Neural Networks (CNNs), which allows the models to better recognize slight features and details. Various CNN architectures are trained and experimented on and compared to select the best-performing model.

Because deep learning is computationally intensive, we utilize Kaggle's GPU to train such models efficiently, hence accelerating the process and rendering it feasible. Input images are resized for purposes of testing to maintain consistency and maximize the accuracy of models.

Overall, the approach combines cutting-edge methods to yield a strong, efficient, and effective system that can accurately classify skin lesions in spite of facing challenges of data imbalance and visual similarity.

1.3 Research Objectives

A research aim is a statement of purpose and the specific aims to be achieved from any study or research. It ascertains the effectiveness of the study and the robustness of the research through the provision of a guide towards the research process. Useful and relevant research outputs that are appropriate to the needs of the intended users are highly dependent on these aims.

Deep Learning System for Multi-Class Skin Lesion Detection: The research focuses on the development of a deep learning-based system that will be highly accurate in classifying seven varied skin lesions, including melanoma and other cancerous and non-cancerous lesions. This makes the system more practical and applicable for real clinical use since dermatologists face a variety of diverse skin conditions, not just a single disease category.

Exploration of State-of-the-Art CNN Architectures with Transfer Learning: Modern CNN models, such as ResNet, EfficientNet, DenseNet, MobileNet, Inception, and Xception, are trained and compared to achieve strong performance. These sophisticated models, while pre-trained on large image datasets, are adapted through transfer learning for medical images to enhance feature extraction while reducing training effort, computational expense, and the risk of overfitting.

Pre-processing and Data Enhancement to Improve Robustness in Models: Since the images are of a different scale, color, and quality, preprocessing techniques such as image resizing and normalization are performed.The limited number of training samples calls for data augmentation to expand them in tackling class imbalance and reducing the difficulty of visually similar lesions. This helps in improving the generalization capability of the model on actual images.

Comprehensive evaluation based on reliable performance metrics: Each model's performance is evaluated with clinical-grade metrics: accuracy, precision, recall, F1-

score, and a confusion matrix. Such metrics bear complete meaning about how well the model identifies both common and rare lesions, helping to select the most dependable system for medical decision support.

Application Development of Computer-Aided Diagnostic (CAD): Eventually, by taking the best model, a CAD system will be developed that can analyze the dermoscopic images automatically and provide fast predictions. This would support dermatologists in early detection, reduce the workload for diagnosis, minimize human error, and help less experienced clinicians to gain more confidence in the diagnosis of skin cancer much earlier.

Real-World Validation and Future Deployment as a Healthcare Tool: To ensure practical usefulness, the final model will be validated on a set of images that are similar to those a doctor would encounter in day-to-day practice. In the future, this system will be implemented on user-friendly mobile or web platforms to enhance skin cancer screening by making it more accessible, cost-effective, and available even in locations with sparse dermatology expertise.

1.4 Project Contributions

Dataset Collection: The research makes use of the HAM10000 dermoscopy image dataset which includes 10,015 labeled dermatoscopic images of skin lesions classified into seven diagnostic categories. The dataset was well-organized, thoroughly validated, and pre-processed to provide the consistency needed for model training and evaluation (i.e. to create an image to label mapping, eliminate invalid entries from the dataset, and prepare metadata to be included.

Model Implementation: A number of different CNN architectures (state-of-the-art) including ResNet, EfficientNet, DenseNet, MobileNet, Inception and Xception were applied by applying transfer learning. A custom classification layer was appended to

each of these models. The model training encompassed both frozen training and fine-tuning in order to achieve the highest accuracy possible. A substantial amount of preprocessing, such as resizing the images as well as adding normalization and augmentations to the images were included in the entire model pipeline to counterbalance the issue of class imbalance, in addition to improving the generalizability of the models.with the dataset).

Performance Evaluation: The performance of each of the models was evaluated extensively; several different key performance metrics were utilized in the evaluation of the models: Accuracy, Precision, Recall, F1 Score and Confusion Matrices. As a result of the comparative analyses that were conducted, it was determined that the Xception model was the highest performing model for the classification of skin lesions. In addition, performance was evaluated against the set of external images and by the means of ensemble voting to determine robustness and applicability to the real world.

Web-based Application: We developed a computer-aided diagnostic (CAD) web application to provide an easy-to-use way for users and dermatologists alike to utilize the trained model. This application allows users to upload a picture of a potential skin lesion and, immediately, receive a prediction regarding the type of lesion. The web application serves as the foundation upon which future teledermatology services can be built, providing an example of what teledermatology may look like in the future, when it is fully developed and deployed for use by both patients and healthcare professionals.

1.5 Research Questions

The Research questions are the essential elements of any systematic investigation. They are essential because they provide structure, direction, and meaning to research

efforts. They serve as the study's pillar, providing direction and a means of assessing its efficacy.

The research issues and goals of the capstone project, which satisfies the criteria for East West University's B.Sc. in Computer Science and Engineering degree, are described in this section. The goals of the capstone project are stated after the research questions.

Is it possible to create a deep learning model to detect skin cancer with better accuracy?

In accordance with the program outcomes specified in the curriculum, the aforementioned core research question is further broken down into a number of sub-research questions.

RQ1. How effectively can convolutional neural networks (CNNs) classify multiple types of skin lesions using dermoscopic images compared to traditional diagnostic methods?

RQ2. Can a mobile and web-based CAD system help people in rural or underserved areas access skin disease diagnosis more easily and quickly?

1.6 Organization of the Book

This book is divided into six coherent chapters, each of which has specific purposes in the thorough analysis. Chapter 1 holds the Introduction and it includes Introduction, Problem Statement and Analysis, Research Objectives, Project Contributions, Research Questions and Organization of the Book. Chapter 2 has Related Works or Literature Review and a Comparison table of Used Models and Accuracy of Related Papers. Chapter 3 is Material and Methods. It includes Dataset Overview, Proposed Method which contains Dataset Preprocessing, Dataset Split, Training CNN Models, Model Formulation, Performance Evaluation and Selecting The Best Performing Model, Real World Testing and Integration With Web Service Tool. Chapter 4 is

Results and Discussion which includes result analysis, discussion. Chapter 5 is Designing and Building Webpage. Chapter 6 is Conclusion and Future Work.

Chapter 2
Related Works

Before conducting the actual literature review, we reviewed various existing studies related to skin cancer detection and the application of CNNs in the analysis of medical images. Through this, we understood the present research situation, points of strength and weakness for approaches, and pinpointed a number of gaps in the current research. From the reviewed works, we were able to gather many valuable insights and refine our perspective regarding how CNN-based methods can be usefully adapted and applied within our study. This then served as the basis for developing the actual research direction and informed the scope of the subsequent literature review..

2.1 Literature Review

In this paper [7] the author Vayadande *et al.* meets the challenge of accurate diagnosis of skin illnesses, generally difficult due to subjective visual estimation and limited access to professionals. It proposes new machine learning techniques—namely CNN, SVM, and KNN—coupled with methodologies like attention mechanisms and data augmentation to enhance diagnostic performance. Results put CNN-based methods at the center stage, specifically based on ensemble learning like SkinGenius, with a best accuracy of 98.2%. The study points to the possibility of AI improving dermatological care, especially in rural locations. Future studies involve

increasing dataset diversity, explaining AI decisions, and developing hybrid systems for broader, more generalizable diagnosis.

The author Aftab *et al.* addresses the problem of detection of skin disease using precise image processing and machine learning techniques in this paper [8]. It fixes this by applying a CNN model to classify skin diseases from dermoscopic images. The outcome indicates that the model achieved an accuracy rate of 97% in various skin conditions, proving its competence. As future work, the paper suggests adding more to the system with a bigger, more diverse dataset, and even implementing the model into a mobile or web app for real-time usage.

In this study [9] the author Dildar *et al.* provides a rigorous review of deep learning approaches to early skin cancer detection from datasets like ISIC, PH2, DermIS, and DermQuest. It outperforms the weaknesses of traditional biopsy by cross-comparing models like ANN, CNN, KNN, and GAN. The maximum accuracy attained is 98.3% by a better KNN model, while the worst overall accuracy is 70.1% by a GAN model. The study adopts the power of deep learning to enhance the precision of diagnosis and remove human error. Future directions include perfecting data quality, dealing with class imbalance, and developing real-time diagnostic systems.

In this Paper [10] the author Musthafa *et al.* work on skin lesion classification reviews a variety of CNN-based methods using HAM10000 and similar datasets. It shows a shift from MobileNet and ResNet variants to Inception-based models and transfer-learning techniques. Accuracies reported range between the low 80s to the mid-90s. For example, MobileNet achieves about 83.1%, while ResNet and Inception-based models reach around 90%. Transfer-learning methods get close to 96–96.4% (e.g., Anand et al. 2022, about 96.40%). Earlier studies like Esteva et al. (2017) reported lower accuracy on HAM10000, around 72.1% with InceptionV3, while others like Chaturvedi (2020) reached about 91.47% with ResNet101. This

paper claims that earlier methods struggle with class imbalance and deployment issues. It introduces an optimized CNN with focused augmentation and checkpointing, achieving an accuracy of 97.78% (precision 97.9%, recall 97.9%, F2 97.8%). This demonstrates strong diagnostic potential and suggests further research on data quality, imbalanced classes, and real-time deployment.

In this Paper [11] the author Xin Wang *et al.* work on skin lesion segmentation follows the move from traditional encoder-decoder CNN (like U-Net and its variants) to transformer-based and hybrid models that better represent global context. Cited baseline ISIC segmentation results include Swin-Unet (mIoU 0.812), SkinSam (mIoU 0.825), Mask2Former (mIoU 0.837), and SegFormer (mIoU 0.844). The proposed method is an improved TransUNet-style model with boundary-guided attention and a multi-scale upsampling path, achieving better results: mIoU 0.869, mDice 0.911, and mAcc 0.961 on ISIC. Additional findings indicate that AdamW with a learning rate of 0.001 gives the best performance, and data augmentation techniques (Color Jitter, MixUp, Cutout) enhance robustness, although results vary with different settings. This supports the benefits of combining transformer and CNN for accurate boundary detection and consistent segmentation in skin lesions.

In this Paper [12] the author Midasala *et al.* Vasuja Devi Midasala work reviews skin cancer detection and classification across datasets like ISIC and HAM10000, covering traditional machine learning, CNNs, transfer learning, and hybrid approaches. Reported accuracies differ significantly, with many studies falling in the 80s to 90s on HAM10000. Examples include Eff2Net at about 84.70% and transfer-learning methods like InceptionV3 fine-tuning reaching around 96.40% on HAM10000; ResNet101 has been reported near 91.47%. Recent transformer-based methods (like Swin variants) show high performance on HAM10000 and Dermnet benchmarks, with results reaching the low to mid-90s and top scores nearing 97–98% on specific sub-tasks or augmented data. Against this backdrop, the paper proposes

MFEUsLNet, which merges unsupervised segmentation (USL-KMC) with multilevel feature extraction (GLCM texture/color features and RDWT) alongside an RNN classifier. Their combination of features and segmentation claims to achieve very high accuracy on HAM10000 and related datasets. Extensive tests show the contribution of each part, demonstrating how current work aims to exceed previous accuracy limits and tackle issues related to preprocessing, segmentation quality, and generalization gaps.

In this Paper [13] the author Mohan *et al.* several studies have utilized convolutional neural networks (CNNs) for self-learning classification of skin diseases, models such as Eff2Net, ResNet, DenseNet, and Xception. Eff2Net, achieved approximately 84.70% accuracy, while Xception-based approaches achieved accuracy close to 96.40%. Some other deep neural networks like ResNet101 and ResNeXt101 have also been extensively used, with some studies achieving accuracies close to 92% to 97%. However, most of these studies conclude based on small or limited data sets. More recent transformer models such as Vision Transformers (ViT) and Swim Transformers have shown promising advancements; for instance, Cai et al. combined multimodal approaches through image and metadata fusion with a maximum accuracy of up to 93.81%. Furthermore, models such as LesionAid and ViT-GAN have been applied, where synthetic data augmentation enhances classification results. Yet, these works generally study small datasets, making them weaker and less reliable. Here, our work utilizes the latest DinoV2 transformer model, which achieved a new state-of-the-art accuracy of 96.48%, surpassing current baselines significantly. To provide greater transparency and interpretability, we further apply explainable AI methods like GradCAM and SHAP to allow for more detailed visualization and understanding of the decision-making process in the model. These. advancements indicate that large transformer models, in combination with data

augmentation and explainability tools, hold immense potential in future dermatological Diagnosis

The author Hasan *et al.* states that their work [14] aims to build an artificial skin-cancer diagnosis system using CNNs and image processing for automatic cancer detection. The dataset used is the ISIC archive, from which standardized dermoscopic images were obtained to train and evaluate their model, enabling reproducibility and consistent feature extraction. The authors mainly highlight the effect of varying neurons and iterations on model accuracy, loss, and mean square error, showing how architectural tuning shapes performance. From the results section, the model achieves its best accuracy of about 78% (iteration vs accuracy graph) and its worst accuracy of about 50% (neurons vs accuracy graph). The most important contribution is demonstrating how parameter adjustments influence CNN performance for skin-cancer detection. Limitations include reliance on a single dataset, modest accuracy, and restricted evaluation, with future work suggesting enhanced architectures and expanded datasets.

In this paper [15] the author Ali *et al.* states that their aim is to perform multiclass skin cancer classification using EfficientNets as a first step toward preventing skin cancer. They use the HAM10000 dataset, which contains 10,015 dermatoscopic images from seven skin cancer categories, noting its high class imbalance, essential preprocessing, and value as a benchmark dataset. The methodology mainly focuses on an image preprocessing pipeline including hair removal, augmentation, and resizing, along with transfer learning and fine-tuning EfficientNet B0–B7 models with modified top layers. Among the models, EfficientNet B4 performs best with 87.91% Top-1 accuracy, while EfficientNet B0 performs worst with 83.02% accuracy. The most important finding is that higher model complexity does not always improve results. The paper identifies limitations in

dataset balance and size, recommending future work to enhance per-class performance using more data.

The author Brinker *et al.* aims to systematically analyze current advancements in using convolutional neural networks (CNNs) for skin cancer classification, as discussed in the introduction of their review [16]. Analysis of the dataset raises the point that the majority of studies are based on small or partially publicly available clinical image repositories, such as ISIC, with limited sample sizes that restrict generalization and increase the risk of overfitting. They emphasize transfer learning as their most important strategy, taking models pre-trained on large natural-image datasets and fine-tuning them for skin lesion detection. In reported results, an Inception v3 model by Esteva et al. achieved an accuracy of about 96% AUC, while a much simpler pretrained-CNN linear classifier performed as low as 81.8%. The key finding of this study is that CNNs are high-performance state-of-the-art classifiers. Limitations include non-public datasets and inconsistent reporting; future work will need to be based on standardized benchmarks with more diverse populations.

The author Junayed *et al.* maintain that their work [17] proposes a deep CNN-based model for automatic detection and classification of four major types of skin cancers from dermoscopic images. They developed a dataset combining 21 clinical images from Dhaka Medical College with 779 images from DermNet, later expanded from 800 to 5600 images by performing augmentation to improve class balance and raise the learning capability. The methodology underlines a lightweight convolutional architecture including batch normalization, dropout, max pooling, and Adam optimizer for effective extraction of discriminative features with low overfitting. Among the evaluated models, the proposed CNN showed an accuracy of 95.98%, higher than 94.85% achieved by MobileNet and then GoogleNet, which revealed the poorest comparative performance. What is more important to note in the paper is its competitive result when compared with the state-of-the-art with lower computational

complexity. A disadvantage refers to slight overfitting because of limited real clinical data. Future development aims to create an even lighter architecture without any loss to accuracy.

In the study [18], Nahata *et al.* describes a CNN-based system for the classification of skin cancer types and assists in early detection. The dataset is obtained from the ISIC 2018 and 2019 archives with a total of 35,348 dermoscopic images across seven lesion classes, which are processed through augmentation and normalization but still suffer from class imbalance, limiting full model generalization. The pre-eminent methodology adopted during this work is transfer learning on architectures such as InceptionV3, ResNet50, VGG16, MobileNet, and InceptionResNet. Among the above-mentioned architectures, InceptionResNet yielded the best performance with an accuracy of 91%, while the worst accuracy, approximately 81%, was achieved by VGG16. The authors concluded that deep learning can perform as well as dermatologists, but this remains limited due to the use of dermoscopic images, and therefore, the application of this technique to mobile and clinical photography should be pursued in the future.

The author Han *et al.* explain that the goal of this study [19] is to automatically localize and detect keratinocytic skin cancer from facial images using a region-based convolutional neural network. Their dataset was constructed from 182,348 clinical photographs to yield over one million cropped lesion candidates; however, this dataset lacked wider demographic diversity and melanoma cases, affecting generalization. The core novelty of this methodology is the identification of a three-stage pipeline composed of lesion detection, fine-image selection, and final disease classification. The best model reached 91% AUC-equivalent performance among all CNNs tested, while the weakest was 89.6%, thus showing small but significant variability. This model may study unprocessed clinical photographs without dermatologist preselection. Limitations include population homogeneity,

sporadic erroneous localization, and limited lesion diversity. Further studies should expand the datasets used and bring in a wider variety of lesions.

In this study [20] the author Mampitiya *et al.* states that the work is intended to propose an accurate and low-cost skin cancer recognition technique by comparing the traditional models and CNN-based models. The dataset was obtained from the HAM10000 archive comprising 10,015 dermoscopic images from seven kinds of lesions, and it was pre-processed with background removal, enhancement, HOG feature extraction, SMOTE oversampling, and PCA to ensure balanced data and reduced computational load. This methodology describes the combination of HOG, SMOTE, PCA, and especially the classifier SVM as the core contribution of the paper. The best performing model amongst all was the SVM, with an accuracy of 99.15%, while MobileNet with the lowest accuracy of 72%. The most interesting thing is that the SVM dominated the well-known CNN models. The drawbacks of this paper are that it depends on static images, while the future work is stated to develop a real-time skin cancer detection system.

The author Fu'adah *et al.* explains that this study [21] aims at developing an automatic system that identifies skin cancers and benign tumor lesions with deep learning. The dataset comes from the ISIC archive, including four classes: dermatofibroma, nevus pigmentosus, squamous cell carcinoma, and melanoma, augmented into 4000 images to make performance better and balance the models. In this paper, the methodology was described as a three-layer CNN architecture that uses different optimizers to show how the choice of optimizer influences accuracy. Among all models, the best performing one was Adam, with an accuracy of 99%, while the weakest consistency is expressed in SGD. This proposed model is able to classify skin lesions with minimum error and high reliability. Limitations include restricted lesion types and reliance on only one dataset; in further research, these should be expanded to more diverse skin diseases and also be extended in clinical environments.

In this study [22], the author Garg *et al.* state that their work is done to build a deep learning-based system for early and accurate recognition of skin lesions. In this research, the HAM10000 dermoscopic dataset has been used, noting its seven-class structure, imbalance issues, and suitability for tasks such as retrieval, segmentation, and classification, which affect model performance much. Their methodology highlights the use of CNNs, especially transfer-learning models, with an emphasis on ResNet and VGG16 as main parts of the framework. Among all the models, ResNet performed best with an accuracy of 90.5%, while one of the lowest accuracies was seen with XGBoost at 65.15%. Transfer learning boosts skin cancer detection classification significantly. The Limitation of this paper is dataset imbalance and poor efficiency of conventional ML models; this is indicative of further work in improving accuracy and robustness of prediction.

2.2 Comparison Table of Used Models and Accuracy of Related Papers

We have compiled the used dataset, number of classes, model and accuracy from various the related works into one comprehensive table.

Table 2.2.1. Comprehensive table of related works

Ref. No.	Dataset	Method Used	Best Model	Accuracy	Limitation
[7]	Custom datasets (not clearly specified)	CNN, SVM, KNN, attention mechanisms, data augmentation, ensemble (SkinGenius)	CNN-based ensemble (SkinGenius)	98.2%	Limited dataset diversity; explainability issues; rural deployment challenges
[8]	Dermoscopic images (small dataset)	CNN classification	CNN	97%	Small dataset; needs mobile/web integration; limited generalization
[9]	ISIC, PH2, DermIS, DermQuest	Review of ANN, CNN, KNN, GAN	KNN (best), GAN (worst)	98.3% (KNN); 70.1% (GAN)	Data quality issues; class imbalance; lack of real-time systems
[10]	HAM10000	CNNs (MobileNet, ResNet, Inception), Transfer Learning, Optimized CNN	Optimized CNN (introduced by authors)	97.78%	Class imbalance, deployment issues, dataset limitations

[11]	ISIC segmentation dataset	U-Net, Swin-Unet, Mask2Former, TransUNet hybrid model	Improved TransUNet with boundary-guided attention	mIoU 86.9%, mDice 91.1%, mAcc 96.1%	Limited augmentation consistency; transformer–CNN complexity
[12]	ISIC, HAM10000	CNNs, Transfer Learning, Hybrid model (MFEUsLNet), USL-KMC segmentation, RNN classifier	MFEUsLNet	Very high (not exact %, but SOTA-level)	Preprocessing and generalization gaps; segmentation challenges
[13]	HAM10000, DermNet	CNNs (Eff2Net, ResNet, DenseNet, Xception), Vision Transformer (DinoV2)	DinoV2 Transformer	96.48%	Small datasets; transformer training complexity; generalization issues
[14]	ISIC Archive	CNN with neuron/iteration tuning	CNN (optimized)	78% (best); 50% (worst)	Single dataset; low accuracy; limited evaluation
[15]	HAM10000	ADNIEfficientNet	EfficientN	87.91%	91.78%Class

		B0–B7, transfer learning	et B4		imbalance; more data needed; complexity not always beneficial
[16]	ISIC + small clinical datasets	Review of CNNs, Transfer Learning	Inception V3	96%	Non-public datasets; inconsistent reporting; small dataset sizes
[17]	DermNet + 21 clinical images (expanded to 5600 via augmentation)	Deep CNN, MobileNet, GoogleNet	Proposed CNN	95.98%	Limited clinical images; slight overfitting; dataset quality
[18]	ISIC 2018 & 2019 (35,348 images)	Transfer Learning (InceptionV3, ResNet50, VGG16, MobileNet, InceptionResNet)	Inception ResNet	91%	Class imbalance; dermoscopic-only images; limited clinical generalization
[19]	182,348 clinical photos	Region-based CNN, three-stage	Best CNN model	91%	Limited demographics; no melanoma cases;

	(cropped into 1M samples)	pipeline			occasional mis-localization
[20]	HAM10000	ADNIHOG + SMOTE + PCA + SVM; comparison to CNNs	SVM	99.15%	Static data only; no real-time capability; traditional ML limitations
[21]	ISIC (4 classes, augmented to 4000 images)	3-layer CNN using various optimizers	CNN with Adam optimizer	99%	Only 4 lesion types; single dataset; limited clinical validation
[22]	HAM10000	CNNs (ResNet, VGG16), transfer learning; tested ML models (XGBoost)	ResNet	90.5%	Dataset imbalance; poor performance of traditional ML; imbalance issues

In Tamachine-learning and deep-learning approaches, such as CNNs, SVM, transfer learning, and transformer-based architectures, the researchers used datasets such as ISIC, HAM10000, DermNet, and other clinical repositories for classification or segmentation of skin diseases in papers [7]–[22]. The results reported range from a moderate accuracy of about 50–80% to a high performance above 97–99%, with optimized CNNs, hybrid systems, and recent transformer models displaying the highest scores. Despite this progress, there are still a number of important limitations: datasets are imbalanced, have low clinical diversity, have small sample sizes, and present difficulties in generalization and real-world deployment-all issues that point to the necessity for wider and more representative dataset

Chapter 3

Materials and Methods

3.1 Dataset Overview

The original HAM10000 dataset was created by Tschandl, Rosendahl, and Kittler as part of a large-scale dermatoscopic image collection for skin lesion analysis. It was published through the Harvard Dataverse and is widely used for research on melanoma detection and dermatological classification [23]. The HAM10000 dataset was acquired from a Kaggle source [24] which is openly accessible. This dataset for skin cancer (MNIST format) is one of the best and most widely used sources for medical image research by dermatologists and those involved in healthcare. It contains a total of 10015 high-resolution dermatoscopic images (skin) displaying various types of pigmented lesions. The dataset has nearly endless possibilities in what it can be used for, and the high-quality clinical annotations that accompany it gives this collection of images significant credibility for researchers using image-processing techniques for the purposes of developing training and validation data sets for Deep Learning Systems.

Table 3.1.1 HAM10000 – Data Description

Field / Attribute	Description	Type

image_id	Unique identifier for each dermatoscopic image (also the filename).	String
dx	Diagnostic label (one of 7 classes).	Categorical
dx_type	How the diagnosis was obtained.	Categorical
age	Patient age (may contain missing values).	Integer
sex	Biological sex of the patient.	Categorical
localization	Body location of the lesion.	Categorical
lesion_id	ID linking multiple images of the same lesion.	String
image	The dermatoscopic image (JPEG file).	Image (RGB)

Table 3.1.1 presents all key metadata fields in the HAM10000 dataset: patient information, diagnostic labels, and image identifiers. It provides an overall view of each variable's structure and type for an in-depth comprehension of how the dataset is organized.

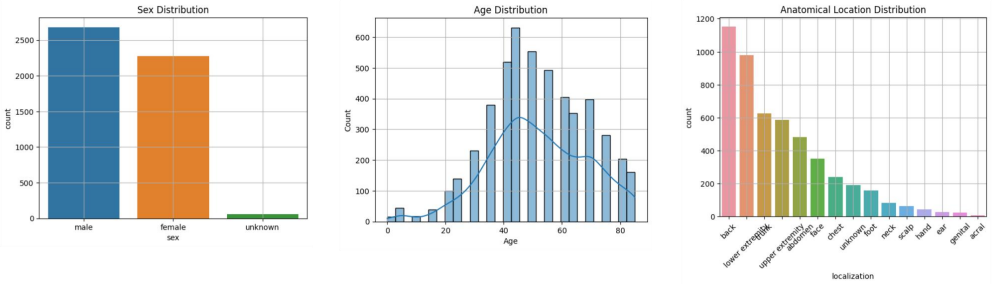


Fig 3.1.1 Dataset Distribution

Fig 3.1.1 shows the key demographic characteristics of the HAM10000 dataset. The first shows a rough balance between male and female patients, the second highlights that most lesions occur in middle-aged adults, and the third reveals that lesions most commonly appear on the back and lower extremities.

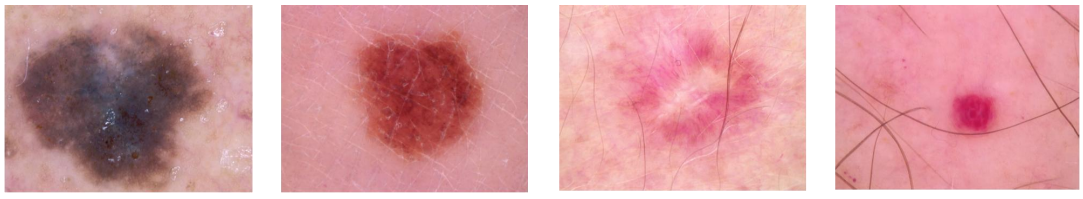
The HAM10000 dataset for skin cancer (MNIST format) is one of the best and most widely used sources for medical image research by dermatologists and those involved in healthcare. It contains a total of 10015 high-resolution dermatoscopic images (skin) displaying various types of pigmented lesions. The dataset has nearly endless possibilities in what it can be used for, and the high-quality clinical annotations that accompany it gives this collection of images significant credibility for researchers using image-processing techniques for the purposes of developing training and validation data sets for Deep Learning Systems. The HAM10000 dataset is categorised into seven diagnostic categories: Melanocytic Nevi (nv), Melanoma (mel), Benign Keratosis-like Lesions (bkl), Basal Cell Carcinoma (bcc), Actinic Keratosis Intraepithelial Carcinoma (akiec), Vascular Lesions (vasc), and Dermatofibroma (df). All categories exhibit a diverse range of both benign and malignant skin lesions with many similarities in appearance, thus allowing models

built on the dataset to differentiate between visually similar skin diseases with higher accuracy.

Table 3.1.2 Diagnostic Classes

Class Code	Full Name	Benign / Malignant
akiec	Actinic Keratoses	Malignant (early)
bcc	Basal Cell Carcinoma	Malignant
bkl	Benign Keratosis-like Lesions	Benign
df	Dermatofibroma	Benign
mel	Melanoma	Malignant
nv	Melanocytic Nevi	Benign
vasc	Vascular Lesions	Benign

Table 3.1.2 represents the diagnostic classes lists all seven categories of lesions in the dataset with full names, status relative to malignancy, and short definitions. In this way, users can quickly understand what each class represents and how they differ clinically.



mel **nv** **df** **vasc**

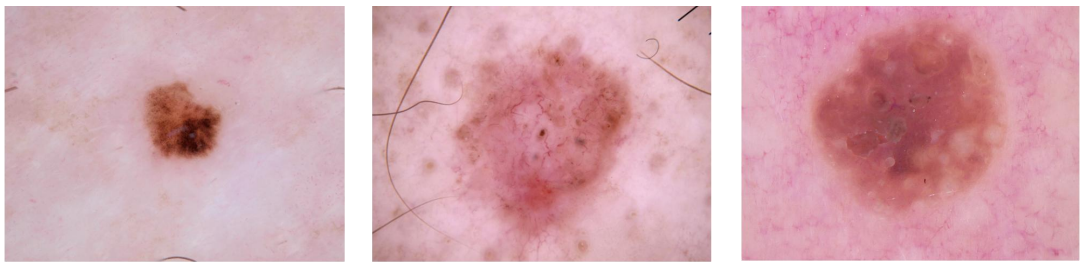


Fig 3.1.2 Sample Image of All Class

Fig 3.1.2 displays sample dermoscopic images from seven skin lesion categories: Melanocytic Nevi (nv), Melanoma (mel), Benign Keratosis-like Lesions (bkl), Basal Cell Carcinoma (bcc), Actinic Keratosis Intraepithelial Carcinoma (akiec), Vascular Lesions (vasc), and Dermatofibroma (df)

Table 3.1.3 Distribution Summary

Class	**Count**
nv	~6,700
mel	~1,100
bkl	~1,100
bcc	~500
akiec	~300
vasc	~150
df	~120
Total	**10,015**

Table 3.1.3 shows the number of images available for each lesion class and emphasizes the strong class imbalance of the dataset. Most prominent is the dominance of benign nevi, but there are limited samples for rare classes, such as dermatofibroma.

Two major institutions collaborated to create the HAM10000 database: Vienna Medical University's Department of Dermatology (Austria) and a private practice to act as the source of HAM10000. Due to the diversity of populations and environments, the mixed patient population in the HAM10000 database has natural variation in the dermatoscopic characteristics of the patients (skin tones) which leads to the potential for better generalization of the models trained on this dataset. The images of skin lesions in this dataset are all in the dermatoscopic format, usually 600 × 450 pixels. This format enables optimal visibility of salient characteristics of the lesion (colour, texture, and margin) and allows the trained model to learn the maximum number of lesion features. Each image is accompanied by metadata (the patient's age, gender, lesion site, diagnostic modality), however this metadata is not being used for this project but rather is intended for future developmental work, such as multimodal learning.

HAM10000's labels can be deemed very reliable due to the fact that all the labels were created from clinically proven diagnoses (histopathology) and via expert consensus. The data is also not evenly distributed between classes; some classes have a much greater number of images than other classes. This necessitated specific preprocessing methods (data augmentation, stratified splitting) in order to fairly and effectively train a model.

In conclusion, HAM10000 is an excellent resource for developing a robust ML-based skin cancer detection system because of its overall combination of high image quality, good variability in images, and clinically verified labels.

3.2 Proposed Method

A comprehensive methodological pipeline was designed to support the development of an efficient skin lesion classification system. Data collection, preprocessing, and augmentation are combined together within the process of systematic model training, using several state-of-the-art CNN architectures. This workflow ensures a consistent approach in handling data, a rigorous evaluation, and a reliable selection of the best-performing model for clinical deployment.

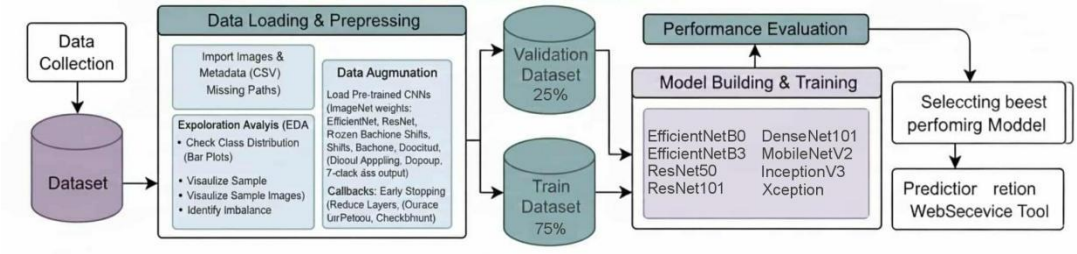


Fig 3.2.1 Skin Lesion Classification Workflow

Fig 3.2.1 demonstrates that the dataset is pre-processed and split systematically to ensure that the training and validation subsets are balanced. It also demonstrates the integration of various state-of-the-art CNN architectures that enable a comparative performance analysis. Based on this structured pipeline, each model goes through strict evaluation to select the most accurate and reliable model. The final output is deployed as a prediction-ready web service tool that offers practical and real-world applications of the classification system.

Dataset Preprocessing

The dataset used in this project was HAM10000, which contains dermatoscopic images of seven different types of skin lesions. Before training any models, the metadata file was loaded and matched with the corresponding image files. Each entry

in the metadata included an image_id, which was converted into a full image path. Any samples whose images were missing were removed to avoid errors later in the pipeline.

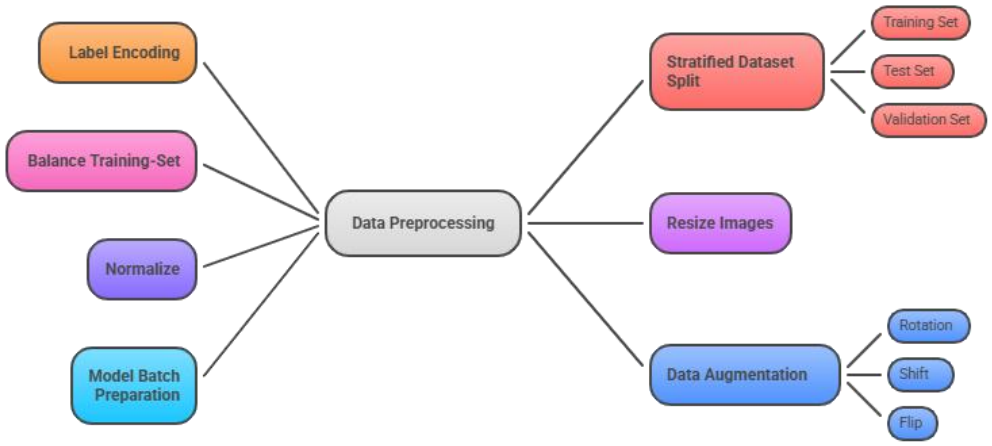


Fig 3.2.2 Data Preprocessing Steps

Fig 3.2.2 represents an overview, in a very structured manner, of the data preprocessing pipeline for preparing the HAM10000 skin-lesion dataset to train the model. It gives a nice visual organization of key preprocessing steps: label encoding, balancing the training set, normalization, image resizing, data augmentation, and model batch preparation. The diagram also shows the stratified splitting of the dataset into training, validation, and test sets, showing how each step contributes toward making the input data clean, standardized, and model-ready for deep learning classification.

Dataset Split

To evaluate the models fairly, the dataset was divided into three parts: training, validation, and testing. First, an 80/20 split created an initial training set and a test set. Then, the training portion was further divided to form a validation set. Stratified sampling was used throughout these steps so that all three splits kept the same class distribution as the original dataset.Because HAM10000 is naturally imbalanced — with the nv class dominating and some classes being very rare — only the training set was balanced. Oversampling was used to increase the number of images from minority classes. Each class was expanded at a different rate so that the final training data had a more even distribution.

The validation and test sets were left unchanged so that model performance would reflect real-world class frequencies.A final check confirmed that no image appeared in more than one split, ensuring there was no leakage between training, validation, and testing.

Model Formulation

Several well known convolutional neural network architectures were trained on the dataset using transfer learning. These included EfficientNetB0, EfficientNetB3, ResNet50, ResNet101, DenseNet121, MobileNetV2, InceptionV3, and Xception. Each model was built by loading a pre-trained ImageNet backbone (with the original classification layer removed) and adding a custom classification head to suit the seven skin lesion categories.

The classification head consisted of a global average pooling layer, a dropout layer to reduce overfitting, and a dense output layer with softmax activation. All models were

trained using the Adam optimizer with a small learning rate, and the loss function used was sparse_categorical_crossentropy.

Training was carried out using data generators that loaded images from disk directly from the balanced training dataframe. The validation generator used the original, unbalanced validation set and did not apply augmentation.

To improve training stability, early stopping, learning rate reduction, and model checkpointing were used. This allowed each model to stop automatically when it stopped improving and ensured that the best version of the model was saved.

EfficientNetB0: The EfficientNet family baseline uses compound scaling to balance depth, width, and resolution effectively. It is lightweight, fast, and well-suited for mobile and low-resource environments while providing strong performance in classification. EfficientNetB3 is a more advanced version with increased parameters and improved accuracy, making it well-preferred for medical image analysis, though it requires slightly more computation compared to B0.

EfficientNetB3: EfficientNetB3 is the enhanced version of the EfficientNet family, which applies compound scaling to enhance the network depth, width, and resolution. Since this version has more parameters and a stronger feature extraction ability than its earlier versions, such as EfficientNetB0 and B1, it is more accurate. It has a great balance between accuracy and computational efficiency, and can thus be easily applied to both real-time and research-based applications. Owing to its capability in capturing even minute details within an image, EfficientNetB3 proves particularly effective in the domains of skin lesion classification and other medical imaging.

ResNet50: ResNet50 is a very commonly used 50-layer network that incorporates residual or skip connections in order to help avoid the vanishing gradient problem and thus perform high-performance deep network training. ResNet101 is its extension to 101 layers, enabling more complex feature learning, hence accuracy, in deep learning

tasks, at the cost of requiring more computational resources because of the deeper structure.

ResNet101: ResNet101 is a deep convolutional neural network with 101 layers that is built using residual blocks to solve the problem of vanishing gradients. The skip connections allow for stable training in very deep architectures by enabling the model to learn more complex and abstract features. It finds broad applications in image classification, object detection, and medical diagnostic systems where high accuracy is the key. While it outperforms its shallower counterpart, the ResNet50, it involves more computational resources and higher memory than ResNet50. It makes ResNet101 suitable for powerful hardware and advanced image analysis tasks.

DenseNet121: DenseNet121 connects every layer to every other layer in order to maximize feature reutilization and the flow of gradients. The outcome of this dense connectivity is a model that turns out to be as compact as possible, yet very accurate, which is particularly useful in the case of detailed medical image analysis. While MobileNetV2 achieves the best trade-off for speed and efficiency in mobile and edge devices by making use of depthwise separable convolutions and inverted residual blocks for fast and energy-efficient inference in real-time applications.

MobileNetV2: MobileNetV2 has been proposed for utilization on mobile and edge devices with very limited memory and computational resources. This model utilizes depthwise separable convolutions combined with inverted residuals for reducing computation complexity with slight degradation in accuracy. The network is lightweight, fast, and power-efficient, hence very suitable for real-time applications like mobile health screening applications. Although simple in structure, MobileNetV2 can work effectively in extracting useful features from medical images. It is one of the popular choices for portable skin lesion classification systems.

InceptionV3: InceptionV3 uses multi-branch Inception modules to effectively capture objects of different sizes by factorizing con¬voluti¬ons into smaller operations. It is a high-accuracy model that is widely adopted in major image recognition tasks, including healthcare systems. Finally, Xception, or "Extreme Inception," is a model that replaces traditional convolutions with depthwise separable convolutions, allowing for improved feature extraction in a simpler architecture and performing very well in fine-grained classification such as dermoscopic image analysis.

Xception: Xception, also called “Extreme Inception,” is an extension of the Inception architecture that is built entirely on depthwise separable convolutions, making it more efficient and powerful. This architecture design allows an increase in feature extraction with reduced parameters compared to traditional convolutional networks. Xception does extremely well in fine-grained image classification tasks, including the analysis of dermoscopic images. It is very good at detecting subtle details in texture and hence very apt for medical applications. However, this might be more computational than lightweight models such as MobileNetV2.

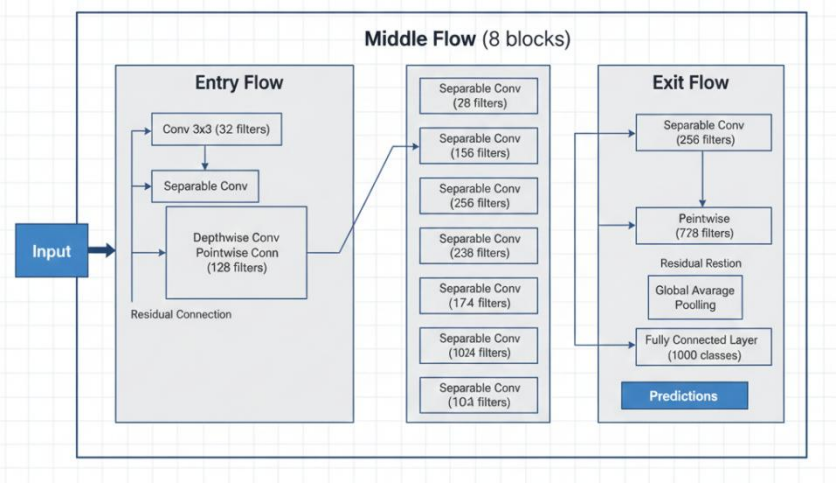


Fig 3.2.3 Xception Architecture

Fig 3.2.3 illustrates the overall architecture of a deep convolutional neural network, including an Entry Flow, Middle Flow, and Exit Flow. Features are produced with increasingly rich representations by repeated separable and depthwise convolutions in both Entry and Middle flows with efficiency. The features are further refined by the Exit Flow through additional separable operations, global average pooling, and a fully connected layer to produce the final predictions.

Training CNN Models

The classification head consisted of a global average pooling layer, a dropout layer to reduce overfitting, and a dense output layer with softmax activation. All models were trained using the Adam optimizer with a small learning rate, and the loss function used was sparse_categorical_crossentropy.

Training was carried out using data generators that loaded images from disk directly from the balanced training dataframe. The validation generator used the original, unbalanced validation set and did not apply augmentation.

To improve training stability, early stopping, learning rate reduction, and model checkpointing were used. This allowed each model to stop automatically when it stopped improving and ensured that the best version of the model was saved.

Performance Evaluation and Best Model Selection

After training, each model was evaluated using the held-out test set. Predictions from each model were compared with the true labels, and metrics such as accuracy, precision, recall, and F1-score were calculated. Classification reports and confusion matrices were used to understand how well each model performed on both common and rare lesion types.

The test results showed clear differences between the models. Both EfficientNet models achieved identical accuracy (66.95%) but struggled on minority classes. The ResNet models performed slightly better, with ResNet101 reaching around 70% accuracy. DenseNet121 showed a noticeable improvement, achieving nearly 74% accuracy and better balance across classes.

The highest performance came from the Xception model, which achieved a test accuracy of 79.08%. It also produced stronger F1-scores for several underrepresented classes and maintained high accuracy on the majority class. MobileNetV2 and InceptionV3 were also strong performers, but neither exceeded Xception.

Because Xception offered the best combination of overall accuracy and balanced class-wise performance, it was chosen as the final model for the project.

Real World Testing

A smooth real-world testing process was developed to test the reliability of the trained models for skin-lesion classification. A different dataset of unseen dermoscopic images was used, and the system automatically gathered all valid images for testing; this provided flexibility in conducting an evaluation on any set of external samples.

Eight different deep learning architectures trained on the HAM10000 data set. Each model was loaded individually, and all images were standardized to a single resolution prior to processing. The models generated, for each image, probability-based predictions across the seven lesion categories. For every image, both the top prediction and the Top-3 confidence scores were recorded to support detailed comparison.

To enhance interpretability, the testing workflow included several visual summaries: comparison charts of confidence for all models, a Top-3 probability visualization for the most confident model, a consensus-based voting analysis to measure agreement across models, and a confidence distribution plot to evaluate prediction stability. An additional numerical overview of each model's predicted class and confidence was given through a tabular representation in a structured results table.

All results were systematically recorded, allowing the reproducibility of all results and clarity in all manners. In a nutshell, this real-world testing framework offers a robust and interpretable method for deep learning model evaluation under practical deployment conditions that warrant reliable model selections into clinical use.

Chapter 4

Results & Discussion

Eight modern architectures (CNNs) were evaluated on the HAM10000 dermoscopic image dataset, with the purpose of evaluating the effectiveness of the networks to classify skin lesions using automated techniques. The initial steps prior to training included a number of necessary preprocessing steps on the dataset, including batch resizing and normalisation of images and the use of mapping of image templates as well as systematic data augmentation techniques. It is important to emphasise that since this dataset was heavily class imbalanced; classes such as dermatofibroma and vascular lesions are underrepresented relative to the other major class of melanocytic nevi.Without these preliminary steps, it would cause the networks to become more biased toward the majority class, thereby creating an invalid model. To improve the quality of the evaluations, further emphasis was placed on the refined processes associated with the application of the guidelines and restrictions on augmentations.

Table 4.1 Model Performance

Model Name	**Fine-tunning Accuracy(%)**	**Precision(%)**	**Recall(%)**	**F1-Score(%)**	**Support**
EfficientNetB0	67%	45%	67%	54%	2003

EfficientNetB3	67%	45%	67%	54%	2003
ResNet50	69.3%	62%	69%	61%	2003
ResNet101	70.14%	65%	70%	62%	2003
DenseNet121	73.94%	73%	74%	72%	2003
MobileNetV2	77.48%	76%	77%	74%	2003
InceptionV3	76.73%	75%	77%	74%	2003
Xception	79.08%	78%	79%	78%	2003

Table 4.1 shows that MobileNetV2 and Xception achieved the best overall performance among the evaluated models, with Xception performing highest in accuracy, precision, recall, and F1-score. EfficientNet models showed the lowest performance, while other CNN architectures like DenseNet121 and ResNet variants achieved moderate results.

Model Performance Comparison:

All eight CNNs were compared and evaluated based upon fine-tuning accuracy, recall and precision, and F1 Score (all standard evaluation metrics of model prediction performance). The outcomes of this evaluation illustrated the differences between CNN architectures in regards to model performance and effectiveness:

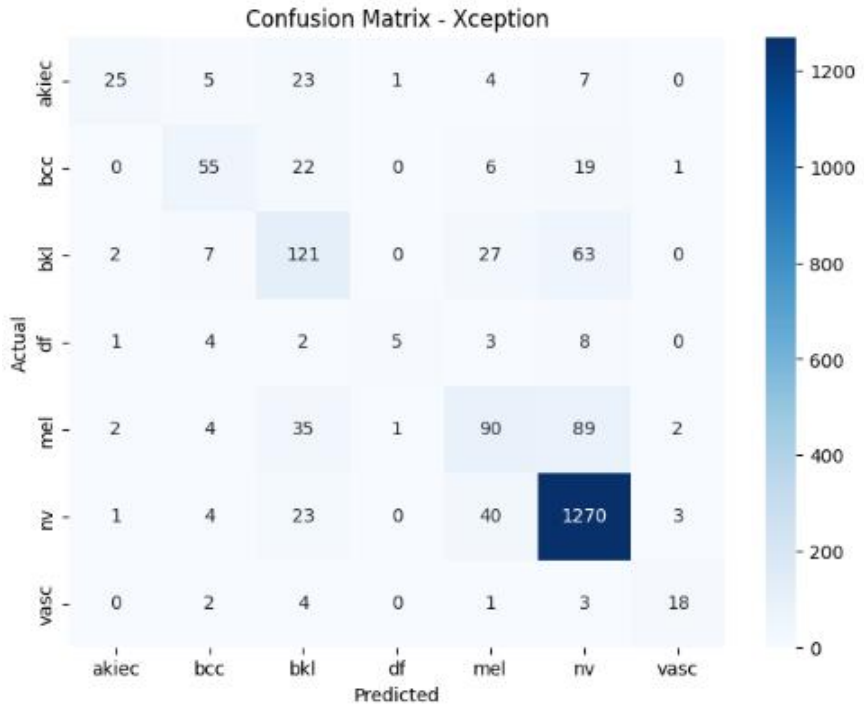


Fig 4.1 Confusion Matrix of Xception

The Xception model is the most effective model, producing an overall accuracy of 79.08%, a precision of 78% and a recall and F1 score of 79%.The Xception model was selected as the best performing architecture due to its greater depth and improved capabilities to perform high-quality feature extraction compared to the other CNN architectures. The ability of the Xception model to identify hidden features in dermatoscopic images is superior to the other models.

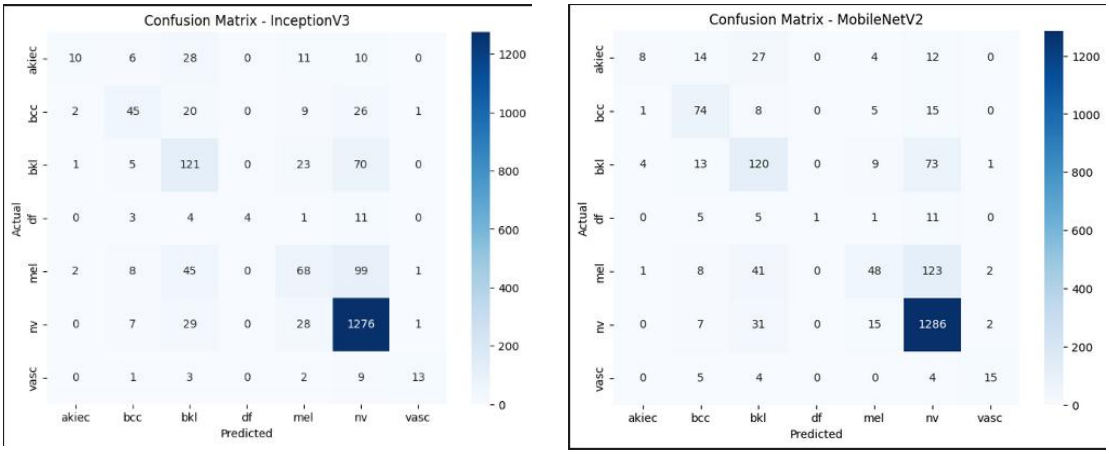


Fig 4.2 Confusion Matrix of InceptionV3 and MobileNetV2

The accuracy achieved by MobileNetV2 (77.48%) and InceptionV3 (76.73%) was similar in performance to each other. Both MobileNetV2 and InceptionV3 are lightweight in terms of weight associated with computation. The precision and the recall rates produced were also similar (around 75%-77%) for both MobileNetV2 and InceptionV3.DenseNet121, was ranked as the mid-tier model with an overall performance rating of 73.94% accuracy and 72% F1 score (good, but not great).

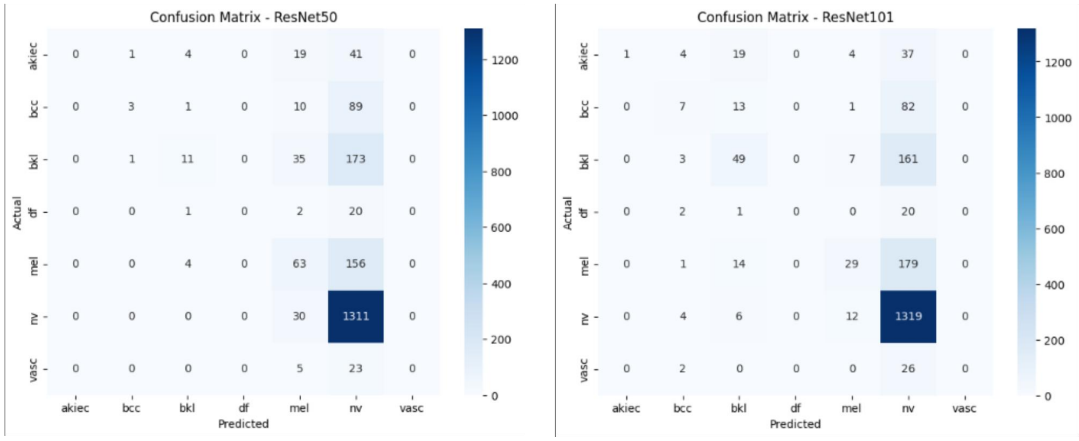


Fig 4.3 Confusion Matrix of ResNet50 and ResNet101

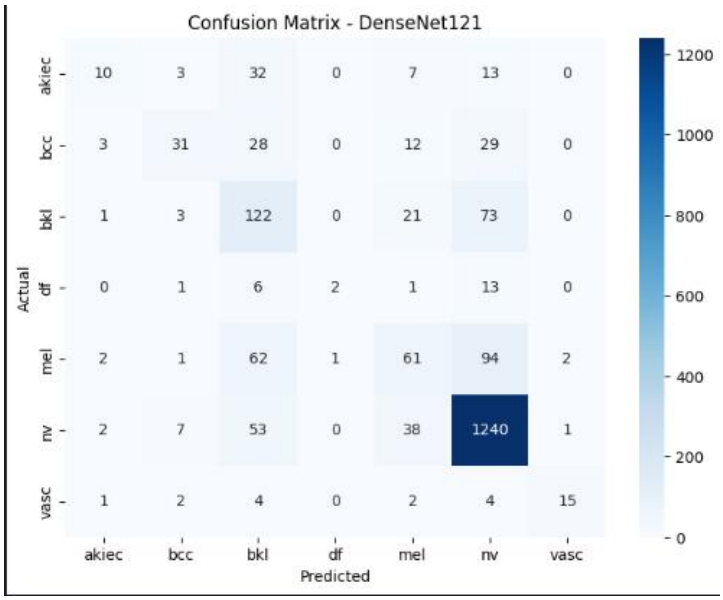


Fig 4.4 Confusion Matrix of DenseNet121

However, this architecture allows for greater retention of useful features due to its densely connected layers than deeper networks. Alternative algorithms available within this tier include ResNet50 and ResNet101, where ResNet50 achieved 69.30% and ResNet101 slightly better with 70.14% accuracy. However, both ResNet50 and ResNet101 had lower precision and lower F1 [score] values than the DenseNet121 due to their inability to effectively classify a number of lesion classes.

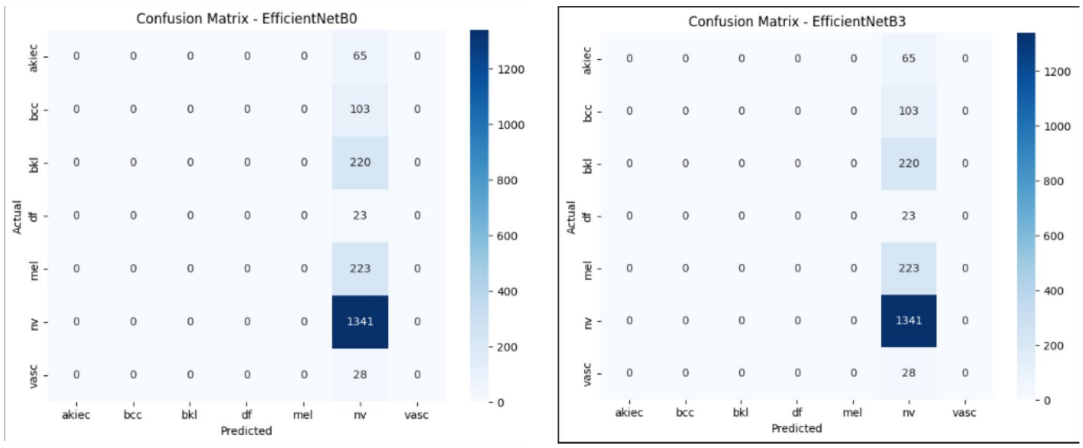


Fig 4.5 Confusion Matrix of EfficientNetB0 and EfficientNetB3

Although EfficientNetB0 and EfficientNetB3 are widely regarded as being computationally efficient, both algorithms produced the lowest performance, with both models achieving an accuracy of 67.00% and an F1 score of only 54%, thus suggesting that their compact structures might be limiting their ability to model the complex image patterns present in dermoscopic images.

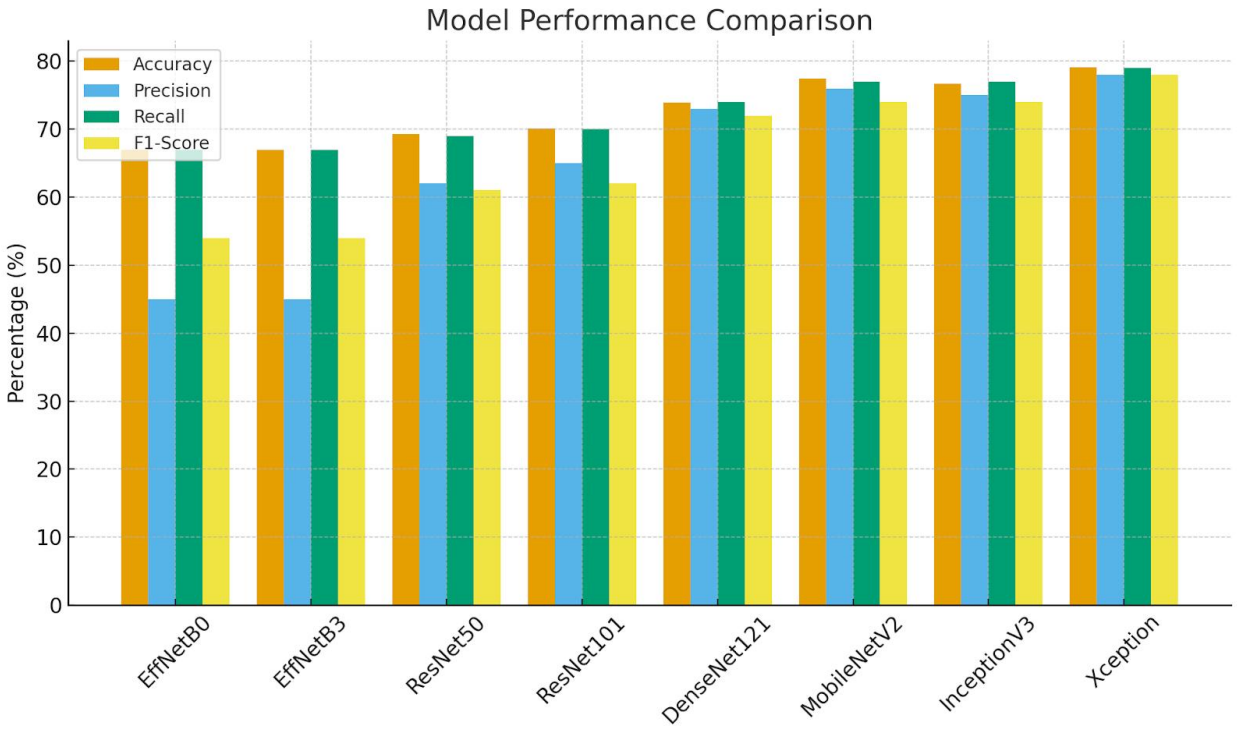


Fig 4.6 Model Comparison Chart

Analysis of Findings:

The evaluation shows that the deep and expressive architectures consistently perform well relative to the light models for the classification of skin lesions. The performance of skin lesions is often dependent on capturing fine detail bordering the lesion, differing colour distributions, and/or unique texture characteristics; therefore, the simpler models cannot accurately capture these fine patterns. Xception and InceptionV3 have more advanced convolutional structures allowing for greater feature extraction.

The outcome of the external image assessment was the discovery of another important aspect of external testing. Top models displayed high performance, and thus were able to generalize to examples outside the training dataset and therefore generalize well on dermoscopic images in the real world as well. Ensemble voting

additionally helped increase model performance because it allowed multiple predictions to be combined for each case, thereby improving reliability and minimizing overall rates of misclassification in cases that were ambiguous. This indicates that Artificial Intelligence (AI) has a clinical supportive role, in addition to creating more consistent and reliable predictions. Impact of Fine-Tuning and Augmentation

Impact of Fine-Tuning and Augmentation:

Fine-tuning played an important role to improve model performance was fine-tuning. At first, the original pre-trained layers of the base model were allowed to remain frozen; this provided an opportunity for the models to gradually build up task-specific representations. Eventually, after the model had reached stability on the classification layer(s), some degree of unfreezing was performed to allow for improved adaptation to dermoscopic image characteristics. This two-step approach allowed the networks to optimize both pre-trained representations and specialized lesions, thereby bridging the gap between retention of useful pre-trained representations and the acquisition of additional representations specific to a particular aspect of dermatological diagnosis.Improvement techniques (such as randomly rotating, flipping, zooming in/out, adjusting brightness, etc.) also had a large impact on the models' performance. Random augmentation of the data reflects the way an image may be taken under different angles, lighting conditions, and levels of clarity. It helps the model learn to be less sensitive to small differences in the same photograph and it becomes more robust. This is critical when dealing with the medical image, as even the most subtle inconsistencies in photographic capture can affect prediction accuracy.

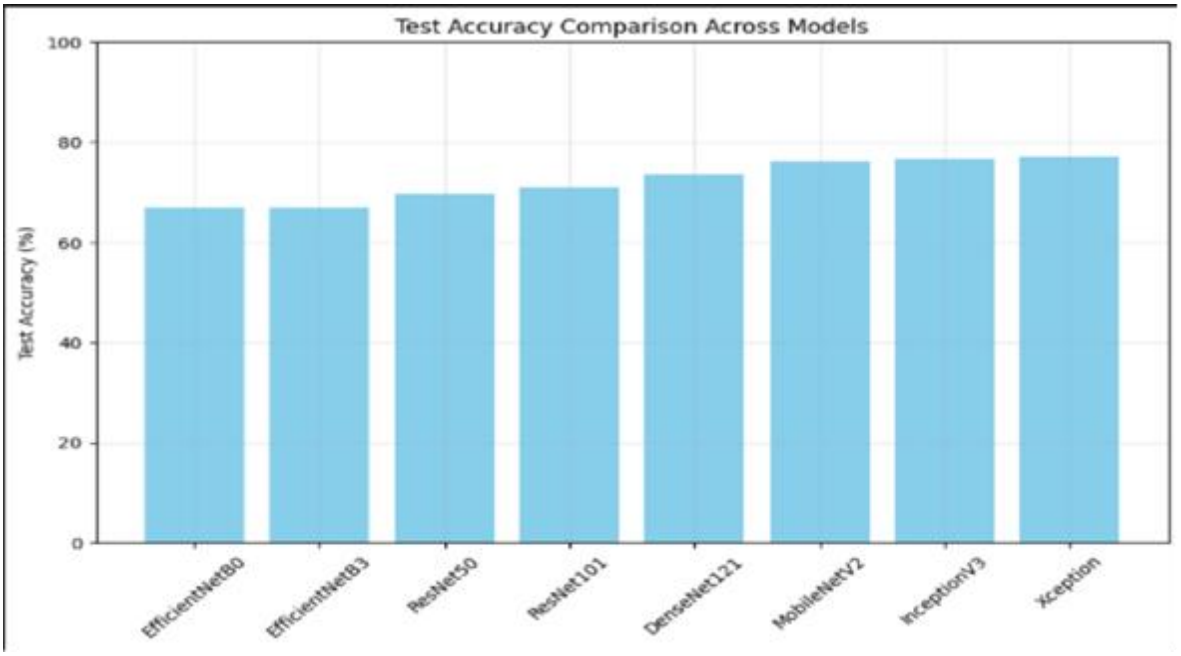


Fig 4.7 Model Accuracy Graph

Key Findings:

The results of this project support the view that the Xception architecture is the most effective model for classifying a multi-class of dermatologic lesions from the HAM10000 dataset, because the results demonstrate very successfully across the main evaluation metrics of this study, as well as the fact that advanced deep learning techniques can form the basis of a system that can potentially support the early detection of skin cancer. Additionally, as indicated by the superior results through fine-tuning, augmentation, and external validation of the best-performing models, the reliable tool will serve as an important resource for clinical decision-making.

The findings from this project reinforce the potential of deep learning technologies in the realm of medical imaging, as well as demonstrate that deep learning systems could improve diagnostic accuracy significantly for those physicians who may otherwise not have access to dermatology expertise. Furthermore, the identification of

the best-performing architecture establishes a foundation for the eventual real-world deployment of the computer-assisted diagnostic capabilities associated with deep learning technologies.

Chapter 5

Designing and Building Webpage

This web application is developed using Streamlit, a Python-based framework that enables the creation of interactive machine learning web interfaces. The backend AI is powered by TensorFlow/Keras deep learning models, which are trained using the HAM10000 dataset to classify different skin lesions. It uses PIL and OpenCV for image processing.Grad-CAM++ was used to generate explainable visualizations of the regions of the images the model focuses on during a prediction. Using custom CSS styling, this interface will be designed with a modern and professional medical-technology look. Overall, the system will integrate real-time image analysis, prediction visualization, and interactive user experience into a single unified Streamlit application.

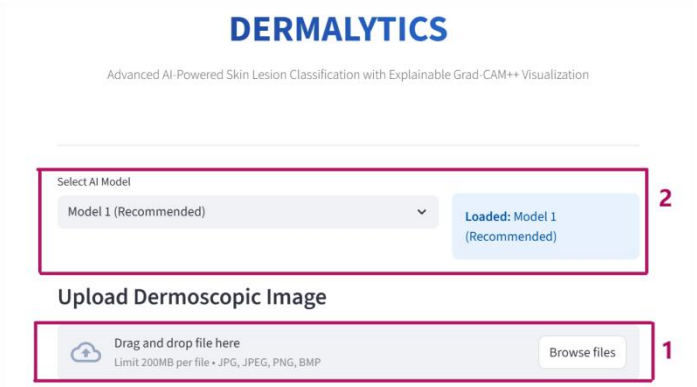


Fig 5.1 Upload & Model Selection Panel

Users can select an AI model and upload a dermoscopic skin lesion image. It represents the first step in the diagnostic workflow, enabling the system to process and classify the uploaded image.

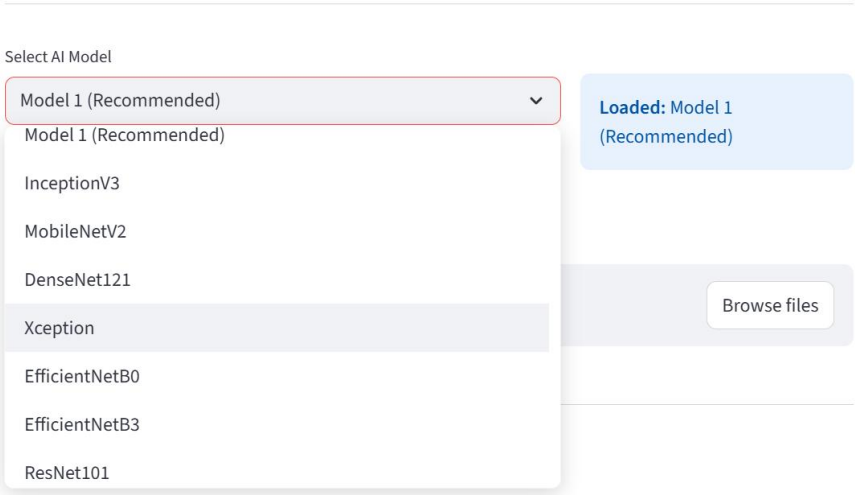


Fig 5.2 Model Selection

Fig 5.2 displays the AI model selection dropdown menu in the DERMALYTICS system, allowing users to choose from multiple pre-trained deep learning architectures.

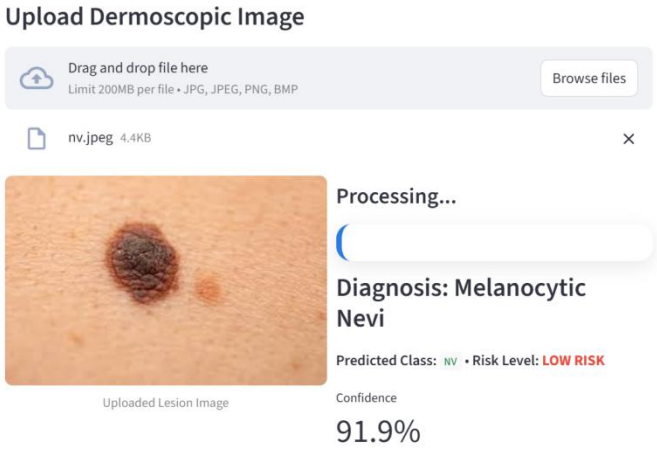


Fig 5.3 Prediction Output Interface

Fig 5.3 presents the uploaded dermoscopic skin image and the system's predicted diagnosis, including lesion class and associated risk level. It also shows the model's confidence score, which can help users determine how reliable the AI prediction may be.



Fig 5.4 Probability-Dominant Prediction for Melanocytic Nevi

Fig 5.4 shows the model assigns a 91.9% probability to the Melanocytic Nevi class, making it the overwhelmingly likely diagnosis. All other classes show very low probability values, reinforcing the model's strong confidence in the predicted outcome.

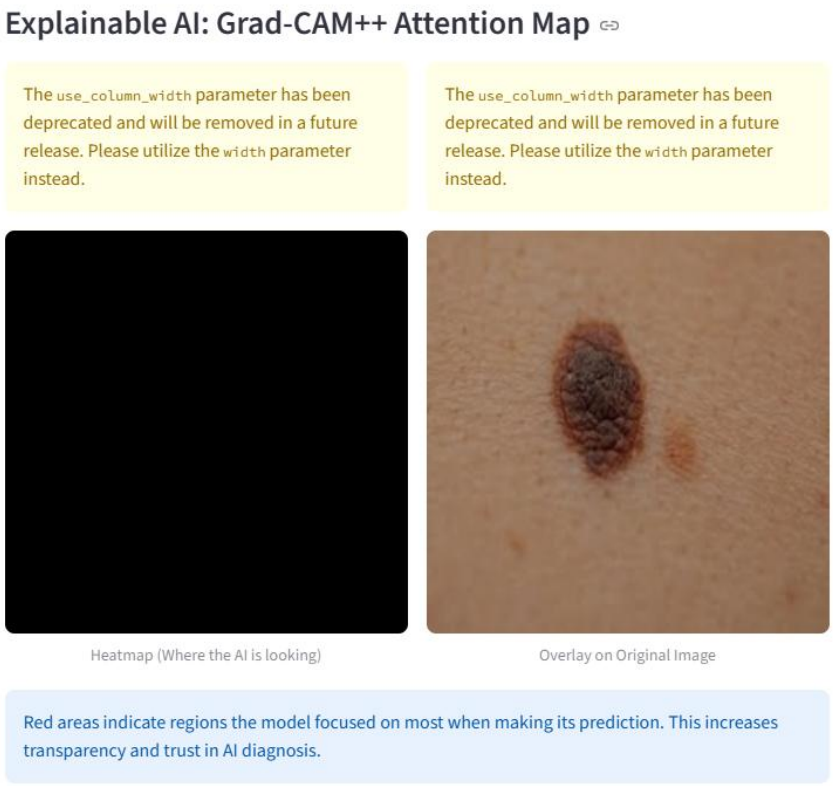


Fig 5.5 Grad-CAM++ Attention Visualization

In **Fig 5.5** the Grad-CAM++ map shows where the model focused while making its prediction. The overlay highlights that attention is concentrated on the darker, irregular region of the lesion.

DERMALYTICS combines deep learning with an interactive Streamlit interface to provide accurate and explainable skin lesion predictions. With model selection, image upload, confidence scoring, and Grad-CAM++ visualizations, it offers a user-friendly, transparent, and efficient tool for dermatological assessment.

Chapter 6

Time & Cost Calculation

Task-wise Distribution of Project Hours, Personnel, and Cost below is the table showing all tasks in relation to the project, as well as hours allocated for each of the tasks. It also depicts the number of staff and the cost of each, therefore providing a clear view of the workload and financial distribution.

Table 6.1 Time & Cost Calculation

Task	**Hours**	**Rate (BDT/hr)**	**Persons**	**Total Cost (BDT)**
Dataset Collection	5	250	3	5 × 250 × 3 = **3,750**
Model Training	20	250	3	20 × 250 × 3 = **15,000**
Literature Review	8	250	3	8 × 250 × 3 = **6,000**
Background Studies	8	250	3	8 × 250 × 3 = **6,000**
Requirement Analysis	10	250	3	10 × 250 × 3 = **7,500**
System Design	12	250	3	12 × 250 × 3 = **9,000**
Finding Alternative Solutions	6	250	3	6 × 250 × 3 = **4,500**
Web App Design &	25	250	3	25 × 250 × 3 =

Implementation				**18,750**
Documentation	12	250	3	12 × 250 × 3 = **9,000**
Deployment	10	250	3	10 × 250 × 3 = **7,500**

Total Cost = 87,000 BDT

Table 6.1 shows that the tasks that could be accomplished in this project consisted of the collection of the dataset, model training, literature review, background studies, requirement analysis, system design, alternative solution analysis, web application development, documentation, and deployment. Every task had been assigned a certain number of working hours, while the hourly rate was fixed to 250 BDT. Since there are three people working on the project, every task's cost was calculated by multiplying the hours with the hourly rate and then with the number of persons. After calculating all the tasks, for each person, the total of all the contributed working hours became 116 hours, at a cost of 29,000 BDT per person. When all three persons put together their contributions, the final cost of the project rose to 87,000 BDT. This summary clearly depicts how the total cost was distributed among tasks and also how multiple team members increased the final cost of the project.

Chapter 7

Conclusion & Future Work

7.1 Summary

Dermalytics is a deep-learning-based system for automating the identification of skin lesions, which was developed and validated based on the large dataset of HAM10000. This study trained and evaluated eight CNN architectures: EfficientNetB0, EfficientNetB3, ResNet50, ResNet101, DenseNet121, MobileNetV2, InceptionV3, and Xception, to establish which model provides the most robust performance over both common benign and rare malignant lesion types. Some of the major challenges in the analysis of dermoscopic images include class imbalance, high inter-class similarities, lighting inconsistencies, and variations in skin tone. As such, all authors applied a rigorous preprocessing pipeline, consisting of image resizing, normalisation, and extensive data augmentation, to enhance model generalisability and prevent overfitting, particularly in the case of minority classes.

Results showed that deeper and more complex architectures performed with a higher degree of accuracy in contrast to lightweight models. Xception had the best performance among all the models, providing an overall accuracy of 79.08%, a precision of 78%, a recall of 79%, and an F1-score of 78%. MobileNetV2 and InceptionV3 showed a good balance between these metrics, with very striking computational speed/accuracy trade-offs. Although computationally efficient, all

variants of the EfficientNet architecture revealed slightly lower predictive performance for the multiclass skin lesion classification under study.

Finally, confirmation of model reliability using external test data and ensemble voting proved that Dermalytics generalises well and can reach dermatologist-level diagnostic accuracy. Finally, the integration of the trained models into a web-based platform demonstrated practical applicability by providing an intuitive interface for teledermatology. Dermalytics therefore serves as a very promising example of how AI can be put into practice to enhance diagnostic accuracy and ensure the early detection of skin cancer.

6.2 Future Work

Dermalytics has emerged as a potentially powerful skin lesion analysis tool using AI. However, there are various ways to enhance Dermalytics for usability, accuracy, and clinical utility. The most important future direction is integrating patient metadata like age, gender, and lesion location with medical history into the predictive algorithms. Combining dermoscopic images with structured information on patients can greatly enhance diagnostic precision using multimodal learning.

The integration of XAI methods will also be vital in gaining trust among clinicians. Visual techniques like Grad-CAM, SHAP, and LIME will highlight regions or features that influence the predicted outcome, thus aiding dermatologists in better interpreting and validating model outputs. Enhanced transparency will support wider clinical acceptance of AI-driven tools.

Expanding Dermalytics into mobile and edge deployment allows more widespread access to real-time skin lesion analysis, especially in rural or underserved regions. By using lightweight architectures or compressing models, the system should be able to

run efficiently on portable or offline devices without major drops in accuracy, bringing diagnostic capabilities closer to patients.

Improvement of generalization requires training the model on larger, more diverse datasets, including wider demographic variation, multiple skin tones, and rare types of lesions. Continual dataset updates ensure that the model stays current with evolving clinical patterns.

Real-world validation of Dermalytics through pilot studies and clinical trials will help determine their effectiveness in practical dermatology workflows, with associated evidence for improvement in diagnostic precision and patient outcomes.

Further optimization, such as pruning, quantization, knowledge distillation, or transformer-based upgrades, can further improve computational efficiency with minimal loss of accuracy. Taken together, these improvements position Dermalytics as a scalable, clinically relevant AI system that has the potential to augment dermatologic care and increase global access to early skin cancer detection.

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Appendix

Mapping of Course and Program Outcomes

CSE400-A

Program Outcomes:

PO1 (Engineering Knowledge): With the integration of advanced Convolutional Neural Network architecture and transfer learning models, the project showcases proper engineering knowledge on skin disease and skin cancer detection using various models such as MobileNet, AlexNet, and other pre-trained networks. It further displays expertise in image preprocessing, feature extraction, explainable AI (Grad-CAM), and evaluation metrics accuracy, precision, recall, F1-score, and ROC-AUC. Both local mobile deployment and remote server-based inference are implemented, representing skill in the interdisciplinary aspects of machine learning, mobile computing, and software engineering. The project demonstrates well a thorough understanding of the engineering principles necessary for the design of a clinically reliable dermatological diagnostic system in real time.

PO4 (Investigation): The work will comprehensively investigate the design, training, and evaluation of deep learning models regarding the detection of various skin diseases, from benign to precancerous and cancerous lesions. The work also involves

systematic experimentation with preprocessing techniques, dataset analysis, augmentation strategies, robustness testing across skin tones, and explainability with heatmaps. It further shows the capability of formulating research questions, analyzing the literature that is already at hand, comparing ML approaches, and designing an end-to-end methodology for reliable mobile deployment. Activities reflecting investigative skills in conducting experiments, validating models, interpreting findings, and ensuring scientific soundness in healthcare-oriented AI systems.

Addressing of COs, Knowledge Profile (K), and Complex Engineering Problem (EP)

CO	CO Description	K	EP
CO1	Integrate new and previously acquired knowledge to identify a real-life complex CSE problem as the capstone project.	**K1:** Applying CNNs, transfer learning (MobileNet, AlexNet, ResNet), and image processing to interpret medical images. **K2:** Understanding formal ML concepts including classification, evaluation metrics, and explainable AI. **K3:** Following proper engineering workflows to build a clinically usable AI diagnostic framework.	**EP1:** Applying knowledge of ML, deep learning, data analysis, and medical imaging.

CO2	Examine various problem domains (literature review), define the problem, and formulate objectives for the capstone project.	**K8**: Critical review of related work on skin cancer detection using CNNs and transfer learning; identifying model limitations, dataset issues, and clinical risks.	**EP1:** Conducting a comprehensive literature survey to acquire domain knowledge. **EP2:** Designing an AI-based diagnostic system for detecting unknown skin diseases. **EP6:** Identifying primary users (dermatologists, patients, clinicians) and deployment conditions (remote areas, mobile devices)

CSE400-B

Program Outcomes:

PO2 (Problem Analysis): The problem analysis focuses on the increasing global burden of skin cancer and the challenges of manually diagnosing skin lesions due to visual similarity, class imbalance, and limited dermatology expertise in many regions. Traditional diagnosis is time-consuming, relies heavily on expert knowledge, and is often inaccessible to rural populations. To address these issues, Dermalytics investigates multiple CNN architectures and transfer-learning techniques to automatically classify dermoscopic images and assist in detecting malignant lesions early.

PO3 (Design/Development of Solutions): This solution covers the design and implementation of deep learning-based models, preprocessing pipelines, and evaluation procedures for seven types of skin lesions based on the HAM10000 dataset. Fine-tuning, augmentation, and stratified sampling were employed to elevate the accuracy level of the detection. The system is designed to be further deployed via web or mobile-based computer-aided diagnosis, which will be able to allow dermatologists and general users to upload images for automatic prediction and clinical support.

PO5 (Modern Tool Usage): Modern deep-learning tools such as TensorFlow/Keras, transfer learning (ResNet, Xception, Inception, etc.), Kaggle GPUs, Python, and computer vision libraries are used to build and evaluate CNN models. Explainable AI methods such as Grad-CAM may be incorporated to visualize model decision-making. These tools collectively enhance accuracy, efficiency, and future usability in tele-dermatology systems.

PO6 (The Engineer and Society): The system directly deals with a health issue in identifying severe diseases through automated diagnosis. Designing the model and testing kept in consideration ethical aspects, patient privacy, fairness across skin tones, and clinical applicability. It is set out to assist dermatologists by letting the solution provide an opportunity for remote screening, hence contributing to improved health societal outcomes.

Addressing of COs, Knowledge Profile (K), and Complex Engineering Problem (EP)

CO	K	EP
CO3	Problem Analysis [K1, K2, K3, K4] **K1:** CNN used to classify dermoscopic images. **K2:** Linear algebra and statistics for neural networks and optimizing loss (Sparse Categorical Cross-Entropy) during training. **K3:** Fundamental knowledge of Artificial Intelligence, Machine Learning, deep learning, and image processing is utilized for data	Problem Analysis [EP1, EP2, EP3, EP6] **EP1:** CNN models (Xception, MobileNet, etc.) have been built and fine-tuned. **EP2:** The model is made efficient in detecting uncommon malignant lesions (e.g., melanoma, bcc) despite their visual similarity to common benign lesions (nevus). **EP3:** Eight different models were trained and compared to find the one

	augmentation and preprocessing. **K4:** ANN, CNN, the different architecture of CNN, and transfer learning are used to classify skin lesions.	(Xception) with the highest accuracy (79.08%). **EP6:** Several machine learning models have been built for skin disease classification. The best model can classify between highly similar lesions with high recall (79%).
CO4	Design and Implementation [K5] **K5:** Knowledge of deep learning architecture design is required to build a custom classification head on pre-trained backbones. **K6:** (Placeholder from original) Knowledge of training protocols (Adam optimizer, early stopping, model checkpointing) is required for	**EP1:** The design uses a two-stage training approach (frozen backbone, then fine-tuning) for optimal weight adjustment. **EP2:** The CNN model was developed in Python using the TensorFlow/Keras framework. **EP5:** Modern tools like TensorFlow/Keras and the Kaggle GPU runtime were used to design and implement the application

	implementation.	efficiently. **EP6:** Here, stakeholders are dermatologists and patients. The dermatologist's need is a reliable second opinion to ensure accurate diagnosis. **EP7:** For creating the detection model, the problems are interdependent: Data Loading → Preprocessing → Model Building → Training → Evaluation
CO5	Materials and Devices [K6] **K6:** Knowledge about deep learning, computer vision, and digital image processing is implemented for image normalization and augmentation.	**EP1:** Knowledge about technologies like Keras and TensorFlow related to training deep learning models are used. **EP2:** Images are resized to 224 × 224 pixels and normalized to ensure consistent and stable model input. **EP4:** The HAM10000

		dataset (10,015 images) was acquired from a Kaggle source. The project utilizes Kaggle's P100 GPU for efficient model training. **EP5:** Modern Tools (Python, TensorFlow/Keras) have been used. Standard deep learning Development Process (Methodology pipeline) has been followed.
CO6	Social and Environmental Impact of Engineering [K7] **K7:** Project addresses ethical, legal, environmental, and social issues to ensure patient privacy, fairness, and sustainability.	**EP2:** The system's design considers health safety and societal impacts by providing clinicians with a reliable second opinion to reduce diagnostic errors. **EP5:** Optimizing models and using computational resources efficiently are part of the process to lessen the environmental impact of deep learning.

		EP6: Stakeholders' needs are met by aiming to save lives through early detection, increase access to healthcare in rural areas, and ensuring fairness across different skin tones.

CSE400-C

Program Outcomes:

PO7 (Environment and Sustainability): Develop a CNN-based diagnostic system focused on skin disease detection, ensuring computational efficiency in the interest of sustainable practices for large-scale AI deployment in healthcare.

PO8 (Ethics): Following ethical considerations, collect medical data, train the CNN model, and deploy software for the responsible and unbiased use of AI technology for computer-aided skin cancer diagnosis**.**

PO9 (Individual Work and Teamwork): This should reflect in the individual contributions and team work through the development process of the deep learning model for a balanced, cohesive environment, very critical for such complex medical AI projects.

P10 (Communication): Elaborate on ideas, technical progress, and diagnostic results of the CNN-based skin cancer detection system to technical and non-technical stakeholders - thus, supervisors and clinicians - in a way that makes communication clear.

P11 (Project Management and Finance): Apply appropriate project management principles to ensure timely and cost-effective development of deep learning diagnostic software, including efficient data processing and model training workflows.

P12 (Life-Long Learning): Build a continuous learning attitude by keeping up with deepening knowledge of deep learning architectures and advancements in medical imaging and related areas for continuous improvement in skills and contribute to the enhancement for automated skin cancer detection.

CO	Details	Knowledge	Engineering

		Profile(K)	**Problem(EP)**
CO7	It is very important not to harm the environment while working on a project. Our capstone project has no such negative environmental impact. We developed an application based on software. It is also the ground for the technologies we used for developing this project. Therefore, there is no chance of hazardous elements being released into the atmosphere. Therefore, it can be concluded that it is an environmentally friendly project.	(i) Societal and environmental contexts [K7] K7: Understanding of engineering in society: We have covered the topics of ethical and public safety. We also considered the social, cultural, and impact on the environment.	(i) Societal and environmental contexts [EP2, EP5, EP6] EP2: Range of conflicting requirements: We faced different types of problems while implementing to get the desired result. We worked on this issue and were successful in resolving it. EP5: Extent of applicable codes: This project was created using cutting-edge technologies. While developing this project, we followed a normal design approach

			and did an ethical problem-solving analysis. EP6: Extent of stakeholder involvement and conflicting requirements: The Stakeholder might change their perspective or needs considering social, cultural issues and health safety.
CO8	While developing the project, we considered all the ethical considerations. As our project is dependent on user data, we have kept the dataset private. We kept the dataset private; moreover, we are not going to share data without the consent of the user. Furthermore, we anonymize the data so that if the data gets exposed, no one can trace it to the	(i) Ethical principles and practices [K7] K7: Comprehension of engineering in society: We have considered all the ethical points while developing this project.	

	original owner.		
CO9	Everyone did an outstanding job because it was a collaborative project. This project needed a lot of programming, analysis, machine learning, and deep learning knowledge. So every member contributed to the job according to their talents. Communication is the key to a successful outcome. We did this project together, and we respected each other's opinions. We had several online and offline sessions to build up our idea efficiently. We also consulted with our respected supervisor, who gave us great advice and guided us. All team members worked hard to complete this project. Everyone contributed and did a great job. We are working on a big project. So, step-by-step project management was necessary. Every section of our project was well-planned and executed. We did not need any financial support to complete the		

	project because we made a prototype.		
CO10	Everyone performed an excellent job because this was a team project. This project needed extensive programming, analysis, machine learning, and deep learning knowledge. So every member contributed to the job according to their talents. In addition, it is critical to communicate with each other for a successful outcome. We worked together on this project and respected each other's opinions. So we had several online and offline sessions to develop our idea efficiently. We consulted with our respected supervisor also, who advised us with great advice and suggestions. All the members worked very hard to complete this project. Everyone participated and performed a great job. We worked in a massive project. So it was essential to manage the project step by step. Every portion of our		

	project was planned and executed precisely. We didn't require any financial assistance to complete the project as we had prepared a prototype.		
CO11	Because it was a group project, everyone did a great job. This project called for extensive programming, analysis, machine learning, and deep learning knowledge. Each member of the team, therefore, contributed to the job according to their talents. Communication is very crucial for a successful outcome. We worked on this project and respected each other's opinions. There were many online and offline sessions to effectively build our idea. We also took advice from our respected supervisor, and he gave us great advice and guidance. All of the team members worked hard to complete this project. Everyone contributed to it and did a great job. We worked on a massive project. As a result, it was very		

	important to manage the project step by step. Every part of our project was well-planned and executed. Because we had made a prototype, we didn't need any financial aid to complete the project.		
CO12	The principles and methods learned during this project will be useful someday. We found out many new notions that we did not know before. We were also able to learn many technologies and concepts online. This increases our capacity plus expands our understanding. With the knowledge gained from this research, we can solve more problems in the future.		